



Starburst Data Response to:

“AI Action Plan”



Corporate Headquarters

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March 15, 2025

Office of Science and Technology Policy (OSTP)
Networking and Information Technology Research and Development (NITRD)
National Coordination Office (NCO)
Attn: Faisal D'Souza
2415 Eisenhower Avenue
Alexandria, VA 22314, USA
Email: ostp-ai-rfi@nitrd.gov

Subject: Response to Request for Information on the Development of an AI Action Plan

Dear Mr. D'Souza,

On behalf of Starburst Data, we appreciate the opportunity to provide input on the development of the AI Action Plan as directed by **Executive Order 14179**. As a leading provider of high-performance data analytics solutions, we are committed to supporting policies that enhance AI innovation while ensuring responsible and ethical AI development.

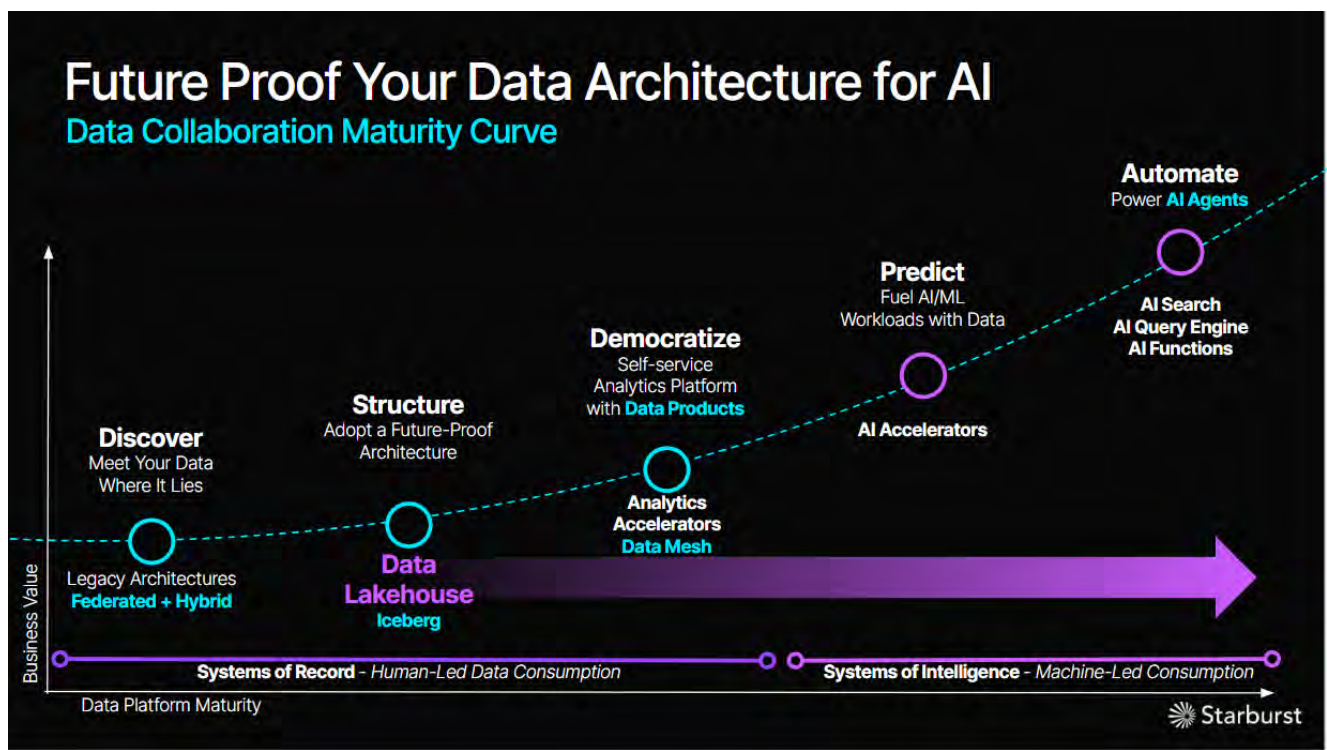
The Starburst Data Core Solution Components

- **Federated Query Engine:** Enables AI applications to access and analyze distributed data in real time without replication.
- **Enterprise Data Governance:** Provides role-based access control, data masking, and encryption to protect sensitive AI training data.
- **High-Performance Data Analytics:** Optimized query execution for AI workloads, reducing latency and improving efficiency.
- **Multi-Cloud & Hybrid Support:** Seamless integration across cloud platforms (AWS, Azure, GCP) and on-premises environments.
- **Cost & Energy Efficiency:** Reduces storage and computing costs by eliminating unnecessary data duplication and movement.

By integrating Starburst Data's federated data solutions into AI infrastructure planning, organizations can achieve AI innovation while maintaining security, efficiency, and regulatory compliance. We look forward to contributing further to the AI Action Plan's development and implementation.

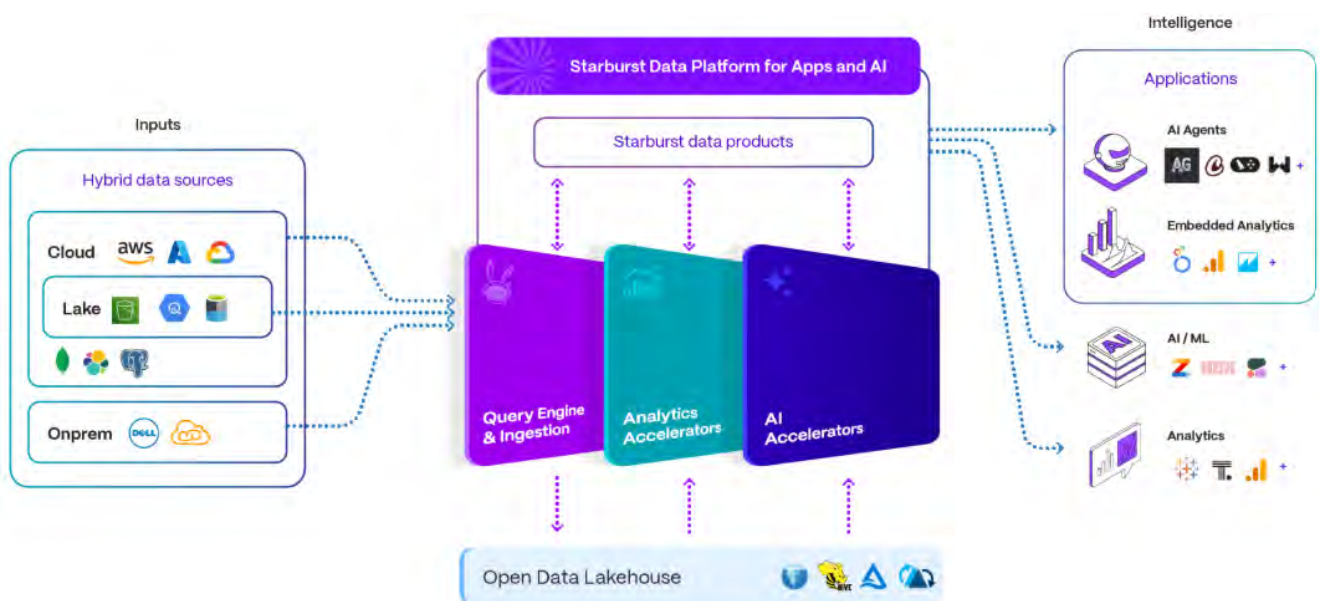
Sincerely,

Seth Nylund
Director, Public Sector
Starburst Data



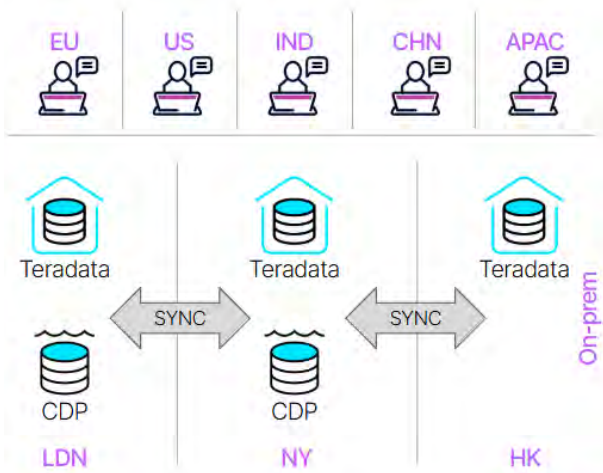
Solution Overview

Starburst Data's technology provides AI-driven organizations with a modern data architecture that removes data silos and enhances AI productivity. Through Starburst's federated query engine, organizations can connect to diverse data sources, reduce infrastructure costs, and accelerate AI-driven insights without compromising security or compliance.



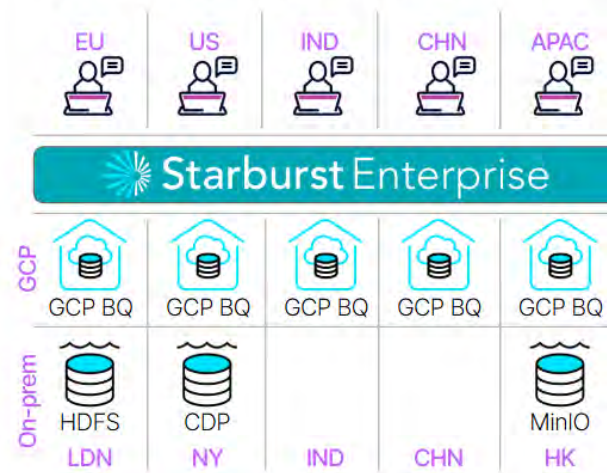
Legacy pre-24

- 5 global regions
- 3 core data platforms/silos



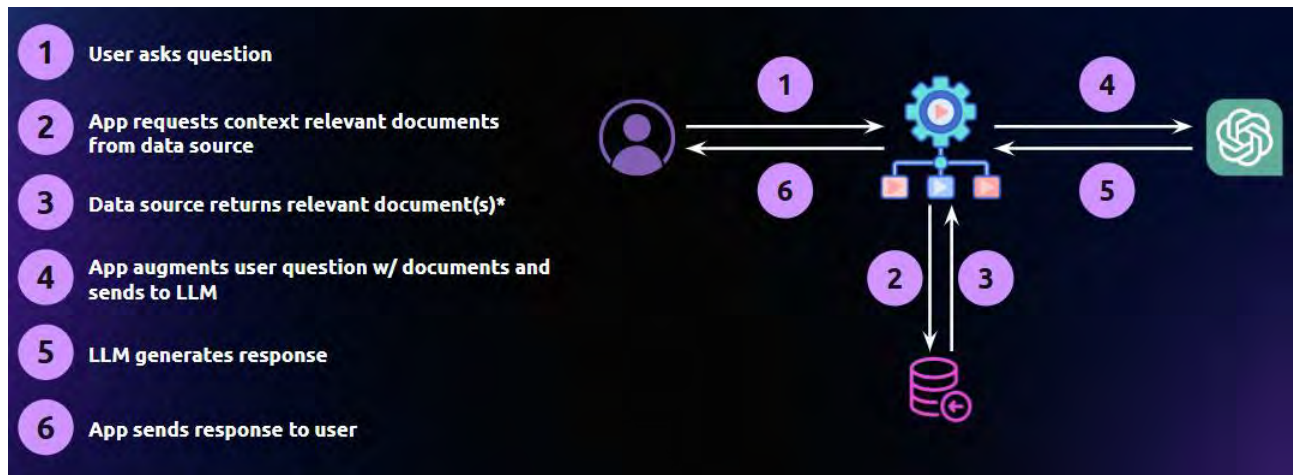
New World post-24

- 5 global regions - Distributed data
- Reduced ETL/ELT
- Single unified (logical) platform



Introduction

Starburst Data provides an open-source, non-proprietary federated query engine that enables organizations to efficiently analyze and manage large-scale, distributed data environments. Our expertise in optimizing data infrastructure uniquely positions us to offer insights into the critical role of data accessibility, security, and governance in AI advancement. Dovetailing with LLM's and LLC's users and analysts of BI tools can now use a "Google Like" prompt to have a single interface to ALL their disparate systems of data.



10 Key Recommendations

1. Data Infrastructure and AI Innovation

- Encourage policies that promote data interoperability and standardization across industries to facilitate AI research and development.
- Support investment in scalable and decentralized data architectures that enable AI models to leverage distributed data sources securely and efficiently.
- Foster initiatives that promote the development of federated learning frameworks, reducing the need for centralized data aggregation while enhancing AI training capabilities.

2. Open Source and AI Model Development

- Promote policies that support open-source AI development, ensuring a collaborative approach to AI advancements without compromising national security.
- Establish public-private partnerships to drive AI innovation while maintaining transparency and accountability.
- Encourage the adoption of explainable AI (XAI) methodologies to enhance trust and reliability in AI-driven decision-making processes.

3. Cybersecurity and AI Data Protection

- Develop guidelines for secure data-sharing protocols to prevent unauthorized access while enabling AI-driven insights.
- Strengthen AI model security against adversarial attacks by investing in research on robust and resilient AI systems.
- Encourage AI governance frameworks that align with zero-trust security principles, ensuring end-to-end protection of AI applications.

4. Regulation and AI Governance

- Ensure AI regulations focus on risk-based approaches that balance innovation with necessary oversight.
- Avoid overly restrictive policies that could hinder AI adoption and competitiveness in the private sector.
- Provide clear and consistent AI compliance guidelines, reducing regulatory uncertainty for businesses investing in AI technologies.

5. AI in Government and Public Sector Applications

- Leverage AI to improve government efficiency and decision-making by integrating AI-driven analytics into public sector operations.
- Establish secure AI frameworks for public-private data collaborations, allowing responsible AI deployment in critical sectors such as healthcare, finance, and national security.
- Promote AI education and workforce development initiatives to equip the next generation with the skills necessary for the AI-driven economy.

6. AI Infrastructure and Compute Expansion

- AI success hinges on access to vast, high-quality datasets and scalable compute power. Starburst enables seamless data access and analytics across distributed environments, eliminating data silos and ensuring that AI models are trained on the most comprehensive and relevant datasets. Our ability to query data wherever it resides reduces the need for costly and energy-intensive data duplication, aligning with modernization and sustainability goals.

7. Energy Efficiency & Modernization

- As AI adoption grows, optimizing infrastructure for energy efficiency is critical. Starburst's federated query engine minimizes data movement and computational overhead, driving down costs and energy consumption. By modernizing data access without requiring massive infrastructure overhauls, we help organizations accelerate AI initiatives in an efficient and sustainable manner.

8. AI Workforce Development

- A thriving AI ecosystem depends on a skilled workforce with access to powerful tools. Starburst empowers data teams by enabling them to access and analyze data quickly, regardless of where it lives. By reducing time spent on complex data engineering tasks, we free up AI and data professionals to focus on model innovation and deployment, helping to close the AI skills gap.

9. AI-Driven Government Efficiency

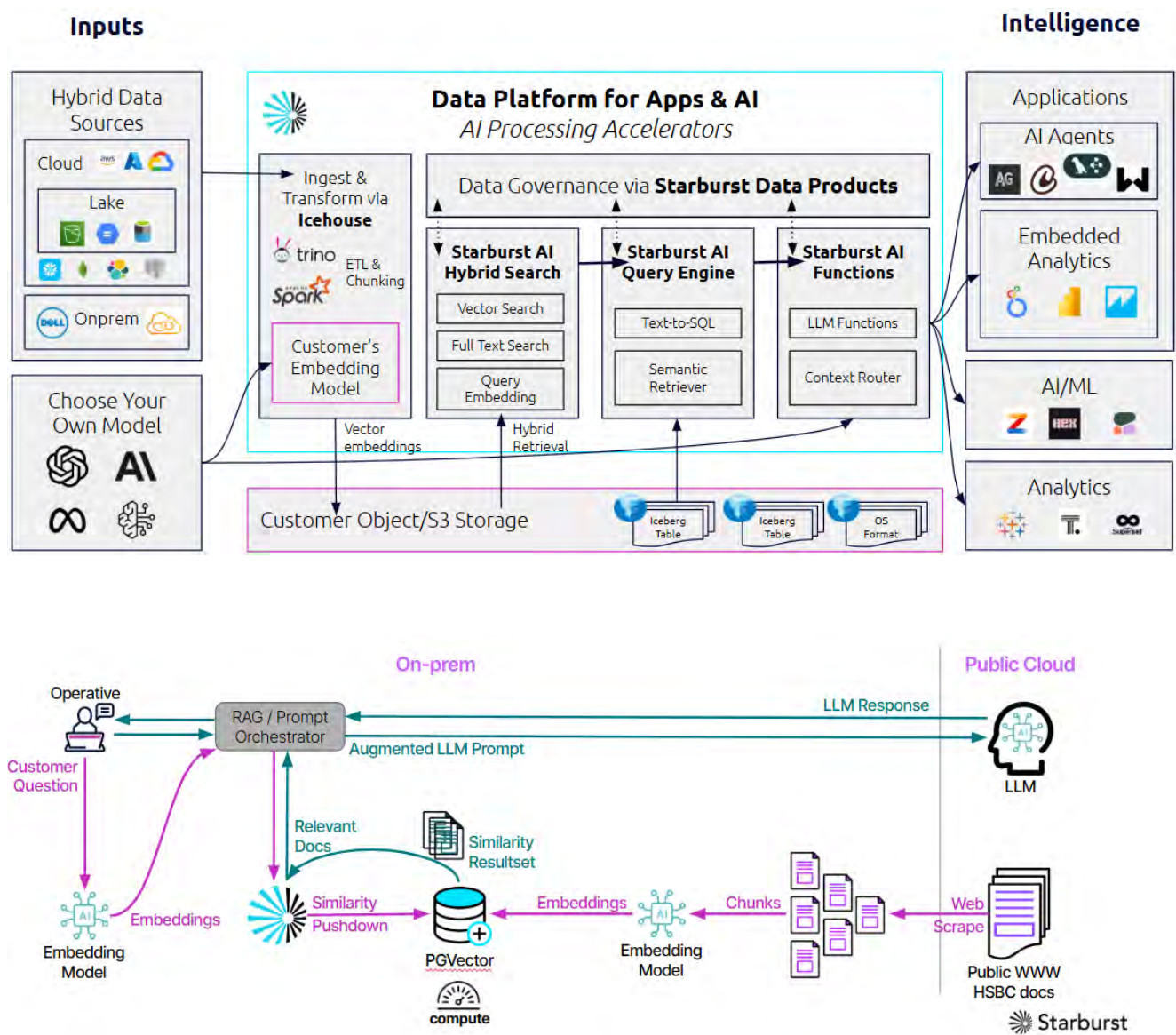
- Starburst's technology accelerates data-driven decision-making across government agencies. By enabling real-time access to federated data, we help agencies leverage AI to enhance operations, improve citizen services, and respond rapidly to emerging challenges. Our solutions provide the agility and scalability required for government modernization efforts.

10. Agile AI Governance

- Responsible AI deployment requires transparent and well-governed data access. Starburst ensures that AI models are built on trusted, well-governed data with full lineage and auditability. Our platform enforces access controls and compliance policies across multi-cloud and on-prem environments, ensuring that AI applications meet the highest standards of security and governance.

Technical Requirements and Solutions

- **Federated Data Access:** Enable AI models to query and analyze data across multiple sources without centralizing or copying data.
- **Scalability & Performance:** Optimize query execution and distributed computing to process large-scale AI workloads efficiently.
- **Security & Compliance:** Enforce data governance, access controls, and security protocols across distributed environments.
- **Open Data Ecosystem:** Support open data formats and interoperability standards for seamless integration with AI and ML frameworks.
- **AI Model Efficiency:** Enhance AI training and inference processes by providing faster, more cost-effective data access.
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Company Information

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Point of Contact

Seth Nylund

Director, Public Sector

Financial & Contracting Information

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Principal Officers

- **Founder & CEO:** Justin Borgman
- **President:** Steve Chung
- **Chief Financial Officer:** Derek Nelson
- **SVP Engineering:** Anders Holden
- **Chief Information Security Officer:** Colton Erickson

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Accelerate mission-critical decisions with secure AI-ready data

Eliminate data silos and unlock AI-powered insights instantly

Solving the most pressing data challenges in government

Government agencies rely on data to make mission-critical decisions. Whether that data is used to feed Business Intelligence (BI) dashboards, data applications, or Artificial Intelligence (AI) models, the goal is the same—clarity of insight.

However, disparate data sources, legacy infrastructure, and evolving security mandates all create obstacles to real-time analysis and faster insights. Adding to this, traditional data access models often require costly and time-consuming data centralization, delaying essential intelligence, slowing operational efficiency, and increasing security risks and compliance issues. Importantly, this is true whether the data involved is used to feed analytics or AI models. In both cases, the project risks stem from data architectural problems.

Starburst eliminates these obstacles by providing a secure, high-performance foundation for all your data—both for analytics and AI. This allows agencies to unify and analyze datasets spread across legacy, cloud, and secure environments without requiring migration, giving them greater flexibility and control over their data initiatives. With improved access to data, agencies can:

- Deliver real-time insights for national security, fraud detection, and government operations by seamlessly accessing all available data wherever it lives.
- Enhance AI models for regulatory compliance, public health monitoring, emergency response, cybersecurity, fraud detection, and defense intelligence by ensuring AI is trained on the most current, comprehensive datasets.
- Reduce operational costs by eliminating redundant data storage and optimizing analytics at scale.
- Accelerate AI-driven decision-making by providing instant access to mission-critical information—without the delays of traditional data pipelines.

Whether supporting regulatory compliance, defense, cybersecurity, fraud investigations, public health initiatives, or emergency response, Starburst empowers agencies to access, analyze, and act on their data instantly—fueling AI-driven innovation while ensuring security and compliance.

The Starburst advantage for public sector

Access data wherever it lives

Government data is often fragmented across multiple agencies, legacy systems, and secure cloud environments. Starburst feeds real-time queries and AI models across data silos—without requiring costly and time-consuming data movement. Agencies can unify data from any structure across cloud, on-premises, and hybrid environments.

Meet security & compliance standards

Government agencies manage vast amounts of sensitive, mission-critical data, requiring strict security, governance, and compliance controls. Starburst enables secure, real-time access across distributed environments, combining fine-grained access control, strong data governance, and a zero-trust model to safeguard data while ensuring AI-driven insights remain accurate, compliant, and efficiently governed across hybrid, cloud, and on-premises systems.

Build the foundation for analytics & AI

Whether detecting fraud, monitoring cybersecurity threats, or optimizing defense logistics, agencies require instant insights from all available data sources. Starburst powers real-time analytics and AI insights by eliminating data bottlenecks, reducing compute time from hours to seconds, and enabling agencies to leverage AI and machine learning models without waiting for data to be centralized.

Adopt a cost-effective & future-proof data architecture

Data duplication, migration, and warehouse dependencies drive up costs and slow down modernization efforts. Starburst eliminates redundant storage expenses and extends the lifespan of legacy systems by allowing agencies to choose whether to query data directly where it resides or centralize it on a modern foundation based on Apache Iceberg—reducing costs while enhancing choice.

Mission-critical public sector use cases

Insider threat detection & mitigation

Starburst analyzes security logs and behavioral data in real time, feeding analytic and AI systems to help detect insider threats. By unifying access to data across different sources, agencies can identify risks early and prevent security breaches.

Fraud detection & prevention

Starburst accelerates fraud detection by enabling real-time, cross-agency data analysis. AI-driven insights help identify suspicious patterns, reduce false positives, and prevent fraud without requiring data movement.

Cybersecurity threat hunting

Starburst unifies threat intelligence, enabling analysts to detect vulnerabilities, track cyber threats, and respond to incidents faster using real-time data across secure networks.

Disaster response & resource allocation

By integrating geospatial, logistics, and sensor data, Starburst helps agencies gather data to help track disaster impact, allocate resources efficiently, and coordinate emergency response in real time.

Regulatory compliance & audit automation

Starburst simplifies compliance by providing federated access to regulatory data, automating audit processes, and reducing the risk of non-compliance penalties.

Citizen service optimization

Starburst enables agencies to analyze data pertaining to citizen interactions and service patterns, improving service delivery, reducing wait times, and enhancing user experience with data-driven decisions.

Infrastructure management & predictive maintenance

Starburst analyzes Internet of Things (IoT) and sensor data to predict infrastructure maintenance needs, preventing failures, reducing costs, and ensuring operational efficiency.

Public health monitoring & response

Starburst provides the foundation for AI-driven insights by integrating public health data, helping agencies track disease outbreaks, allocate resources, and respond effectively to health crises.

Powering AI-driven government insights with Starburst

Access data wherever it lives

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Unified data access

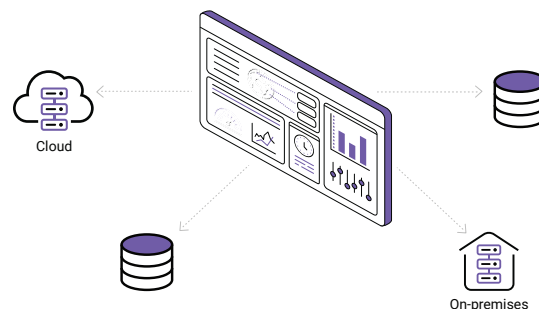
Starburst connects to over 50+ data sources—from legacy on-premises databases to secure cloud environments—allowing agencies to query data instantly without disruption.

Security & compliance built-in

Starburst ensures mission-critical security with fine-grained access controls, end-to-end encryption, and seamless integration into existing cybersecurity and compliance frameworks.

Faster decision-making

Starburst significantly reduces query times from hours to seconds while feeding state-of-the-art AI models. Together, this allows decision-makers to act faster with high-quality data.



About Starburst

Starburst unifies data across clouds and on-premises, accelerating AI innovation with fast access, collaboration, and governance—future-proofing data architecture to fuel innovation with AI at scale.

Learn more at starburst.ai

